

# Yurina Nakazato

## Curriculum Vitae (last updated July 2026)

### Address:

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New York, NY 10010

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**Web:** <https://yurinanakazato.github.io>

### Research Interests

Galaxy formation and evolution, cosmology, first stars and galaxies, cosmological simulations, multi-wavelength observations

### Work Experience

Oct. 2025 - present      **Flatiron Research Fellow, Center for Computational Astrophysics**, Flatiron Institute, New York, US

### Education

Apr. 2023 - Sep. 2025      Ph.D. student, The University of Tokyo, Japan (Completed ahead of schedule)  
Thesis: *Formation and Evolution of High-Redshift Galaxies Using Cosmological Simulations and Multi-Wavelength Observations*  
Advisor: Naoki Yoshida

Apr. 2021 - Mar. 2023      M.S. in Physics, The University of Tokyo, Japan  
Thesis: *Formation and evolution of star clusters and galaxies in the early Universe*  
Advisor: Naoki Yoshida  
President's Award (1/3000), Dean's Award (1/128)

Apr. 2017 - Mar. 2021      B.S. in Physics, The University of Tokyo, Japan

### Fellowships and Awards

Oct. 2025      The L'Oréal-UNESCO For Women in Science Japan Fellowship

Aug. 2025      The Excellence Award for Final Exam

Jan. 2025      Flatiron Research Fellow

Jan. 2025      CITA Fellow (declined)

Feb. 2024      Outstanding Poster Award at the International Symposium on Quantum Electronics 2024

Apr. 2023- Sep.2025      JSR fellowship

Apr. 2023- Sep.2025      Japan Society for Promotion of Science (JSPS) Research Fellow, DC1

Mar. 2023      The School of Science Encouragement Award

Mar. 2023      The University of Tokyo President's Award

Mar. 2023      The Excellence Award for Qualifying Exam

Oct. 2021- Sep.2025      The International Graduate Program for Excellence in Earth-Space Science (IGPEES), the University of Tokyo

Aug. 2021      Best Oral Presentation Award at 51th astronomical meeting for junior researchers

### Leadership & Outreach

Jun. 2023      Organizer of Get-Together Event for Women in STEM, the University of Tokyo

Jul. 2022      Public Talk at Women in STEM, School of Science, the University of Tokyo

Dec. 2019      Rikejo Initiative, the University of Tokyo

Oct. 2019      Invited talk at Dow Chemical Company, Tokyo, Japan

## Grants as PI

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|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Apr. 2023-Mar.2026 | Evolution of first galaxies with multi-wavelength observations and numerical simulations, 4.2M JPY (30K USD), JSPS Grant-in-Aid for Early-Career Scientists, No. 23KJ0728 |
| Apr. 2023-Mar.2026 | Grand Aid from JSR fellowship, 3M JPY (21K USD)                                                                                                                           |

## Coverage in Media

**Hunting Faint Galaxies with JWST as a Test for the Dark Matter Model** based on Williams et al. 2024  
- *Bright galaxies put dark matter to the test*, UCLA

**The core of the most distant galaxy protocluster observed by JWST & ALMA** based on Hashimoto et al. 2023  
- *The Strong Tag Team of the James Webb Space Telescope and ALMA Captures the Core of the Most Distant Galaxy Protocluster*, Kavli IPMU, Tsukuba Univ., Waseda Univ., Nagoya Univ.

## Teaching & Professional Service

### Supervising

|                   |                                                                     |
|-------------------|---------------------------------------------------------------------|
| Jan. 2026-present | co-supervisor of Sho Ebihara (UTokyo & University of Manchester)    |
| Jul. 2026-present | Primary supervisor of Jessica Kelley-Derzon (University of Florida) |

### Teaching

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|----------------------|-------------------------------------------------------------------|
| Apr. 2021-Aug. 2021  | Teaching Assistant of Fluid Mechanics, the University of Tokyo    |
| Apr. 2019- Jan. 2024 | Language Assistant, International Lounge, the University of Tokyo |

### Professional Service

|              |                                                                                                                                   |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------|
| 2026         | <b>Reviewer for Grant Proposals of UK Research and Innovation (UK)</b>                                                            |
| 2026         | <b>Reviewer for Subaru proposals</b>                                                                                              |
| 2025-present | <b>Referee of Astronomy &amp; Astrophysics (A&amp;A), The Astrophysical Journal (ApJ), The Open Journal of Astrophysics (OJA)</b> |
| Mar. 2026    | <b>Chair</b> , Astronomical Society of Japan Meeting (ASJ) (>100 participants)                                                    |
| Mar. 2026    | <b>Chair</b> , Ishigaki ALMA Workshop, 2026 (~50 participants)                                                                    |
| Mar. 2026    | <b>Chair</b> , JWST Tokyo Conference (>200 participants)                                                                          |
| Apr. 2025    | <b>Organizer</b> , One-day workshop on stellar-cluster simulation, The University of Tokyo, Tokyo, Japan                          |
| Dec. 2024    | <b>Organizer</b> , One-day workshop on high-z cosmological simulations, The University of Tokyo, Tokyo, Japan                     |
| Aug. 2023    | <b>Organizer</b> , The 53rd Summer School on Astronomy and Astrophysics, The University of Tokyo, Tokyo, Japan                    |
| Mar. 2023    | <b>Workshop Organizer</b> , Astrophysics Workshop for Young Researchers, The University of Tokyo, Tokyo, Japan                    |

## Proposals

### co-Investigator on JWST proposals

- GO10830, PI: Alejandro Crespo-Gomez  
*Disclosing the nature of N-emitters in the early Universe: Rest-frame nearIR imaging of RXCJ2248-ID at  $z = 6.1$*

### co-Investigator on ALMA proposals

- 2025.1.00666.S, PI: Y.Osone, 14.0h  
*Characterizing the Nature of the Overdensity Region in the Epoch of Reionization with [OII] 145um*
- 2023.1.00843.S, PI: Y.Tamura, 15.6h  
*Disentangling the ionization structure of HII regions in a density-bounded Ly $\alpha$  emitter in the Reionization Era*

3. 2023.1.00193.S, PI: Y. Fudamoto, 17.3h  
*[CII] 158um emission line and dust observation of the most distant known overdensity of galaxies at  $z = 7.88$*
2. 2023.1.00235.S, PI: Y. Fudamoto, 10.0 h  
*The ALMA-JWST synergy: [OIII]88um and [CII]158um emission line observations of  $z = 9.51$  galaxy identified by JWST*
1. 2023.1.01362.S, PI: Y. Tamura, 25.9 h  
*Pandora's ELPIS: The Emission-Line Protocluster Imaging Survey of the furthest overdensity beyond Pandora's Cluster*

### Active collaborations

- Supersonic Project
- FirstLight simulations
- Reionization and the ISM/Stellar Origins with JWST and ALMA (RIOJA, JWST GO1840)

### Selected Talks at Conferences and Workshops

total talks will be listed here

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|-----------|------------------------------------------------------------------------------------------------------------------------------|
| Mar. 2026 | <b>Tokyo-JWST Conference 2026</b> , Tokyo, Japan                                                                             |
| Mar. 2026 | (keynote) <b>Star Forming Galaxies in the Reionization Epoch: ALMA-Driven Studies with the Aid of JWST</b> , Ishigaki, Japan |
| Feb. 2026 | (invited) <b>From Dust till Dawn</b> , Leiden, Netherlands                                                                   |
| Aug. 2025 | (invited) <b>The Fine Structure Line Workshop 2025</b> , Hokkaido, Japan                                                     |
| Sep. 2024 | <b>ILR workshop 2024 @Osaka: From Galaxy Formation to Supermassive Black Holes</b> , Osaka, Japan                            |
| Aug. 2024 | <b>IAUS391</b> , Cape Town, South Africa                                                                                     |
| Jul. 2024 | <b>European Astronomical Society (EAS) Annual meeting 2024</b> , Padova, Italy                                               |
| Nov. 2023 | <b>Resolving the Extragalactic Universe with ALMA &amp; JWST</b> , Waseda Univ., Japan                                       |
| Jul. 2023 | <b>Shedding new light on the first billion years of the Universe</b> , Marseille, France                                     |
| Sep. 2022 | <b>The 9th East Asian Numerical Astrophysics Meeting (EANAM9)</b> , Okinawa, Japan                                           |
| Aug. 2022 | <b>Star Formation in Different Environments (SFDE) 2022</b> , Quy Nhon, Vietnam                                              |
| Jul. 2022 | <b>National Astronomy Meeting (NAM) 2022</b> , online, hosted by the University of Warwick                                   |
| Nov. 2021 | <b>IAU Symposium 362: The predictive power of computational astrophysics as a discovery tool</b> , online                    |

### Seminars (Selected)

total: 19 talks

- |           |                                                                                  |
|-----------|----------------------------------------------------------------------------------|
| Jun. 2026 | <b>University of Massachusetts, Amherst</b> , special seminar, The United States |
| May. 2026 | <b>Princeton University</b> , invited seminar, The United States                 |
| Apr. 2026 | <b>University of Maryland</b> , invited seminar, The United States               |
| Nov. 2025 | <b>Flatiron Institute</b> , CCA-NY area galaxy seminar, The United States        |
| Aug. 2025 | <b>Hokkaido University</b> , invited seminar, Japan                              |
| Jul. 2025 | <b>Kyoto University</b> , invited seminar, Japan                                 |
| Feb. 2025 | <b>Tsukuba University</b> , invited seminar, Japan                               |
| Apr. 2024 | <b>Scuola Normale Superiore</b> , seminar, Italy                                 |
| Dec. 2024 | <b>Universidad Autónoma de Madrid</b> , special seminar, Spain                   |
| Dec. 2024 | <b>Centro de Astrobiología, (CAB, CSIC-INTA)</b> , special seminar, Spain        |
| Feb. 2023 | <b>UCLA</b> , invited seminar, The United States                                 |

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## Working experience as a visiting researcher (student)

- Apr. 2024 - Jul. 2024      **Visiting Student at Scuola Normale Superiore di Pisa**
- Host researcher is Prof. Andrea Ferrara.
  - Worked on radiation-driven dusty outflow at  $z > 10$  galaxies.
- Nov. 2023 - Dec. 2023      **Visiting Student at Universidad Autónoma de Madrid**
- Host researcher is Prof. Daniel Ceverino.
  - Worked on formation and evolution of clumpy galaxies by using FirstLight simulation
- Feb. 2023 - Mar. 2023      **Visiting Student at UCLA**
- Host researcher is Prof. Smadar Naoz.
  - Worked on Supersonic Project [link]

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## Competencies

Scientific programming    : Python, C++, Fortran, Latex  
Simulation software        : AREPO, ART, RAMSES-RT, RAMSES  
Astrophysics tool         : CLOUDY(photoionizaion code), Astrodendro, SKIRT(dust radiative transfer)

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## References

Prof. Naoki Yoshida (Ph.D. supervisor), The University of Tokyo, naoki.yoshida@ipmu.jp  
Prof. Andrea Ferrara, Scuola Normale Superiore, andrea.ferrara@sns.it  
Prof. Smadar Naoz, University of California, Los Angeles, snaoz@astro.ucla.edu  
Prof. Daniel Ceverino, Universidad Autonoma de Madrid, Madrid, daniel.ceverino@uam.es  
Prof. Rachel Somerville, Flatiron Institute, New York, rsomerville@flatironinstitute.org